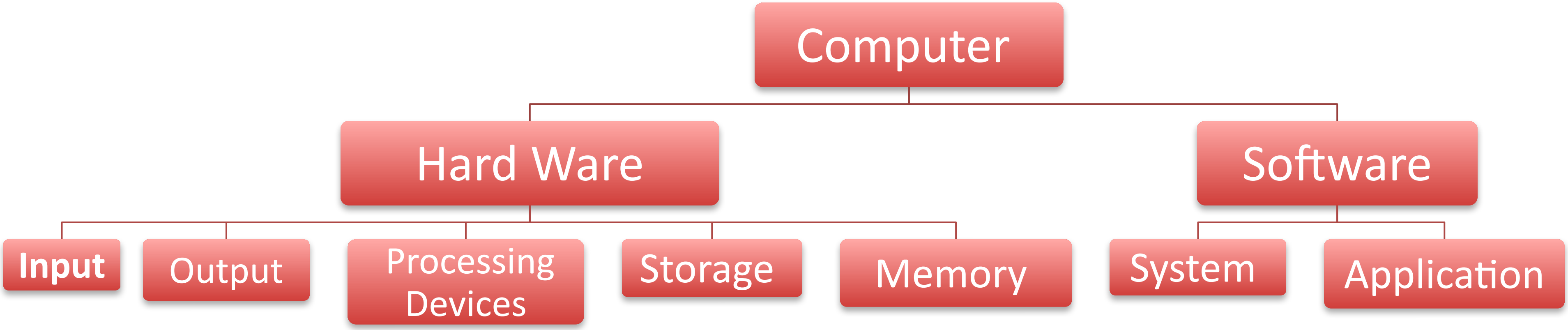
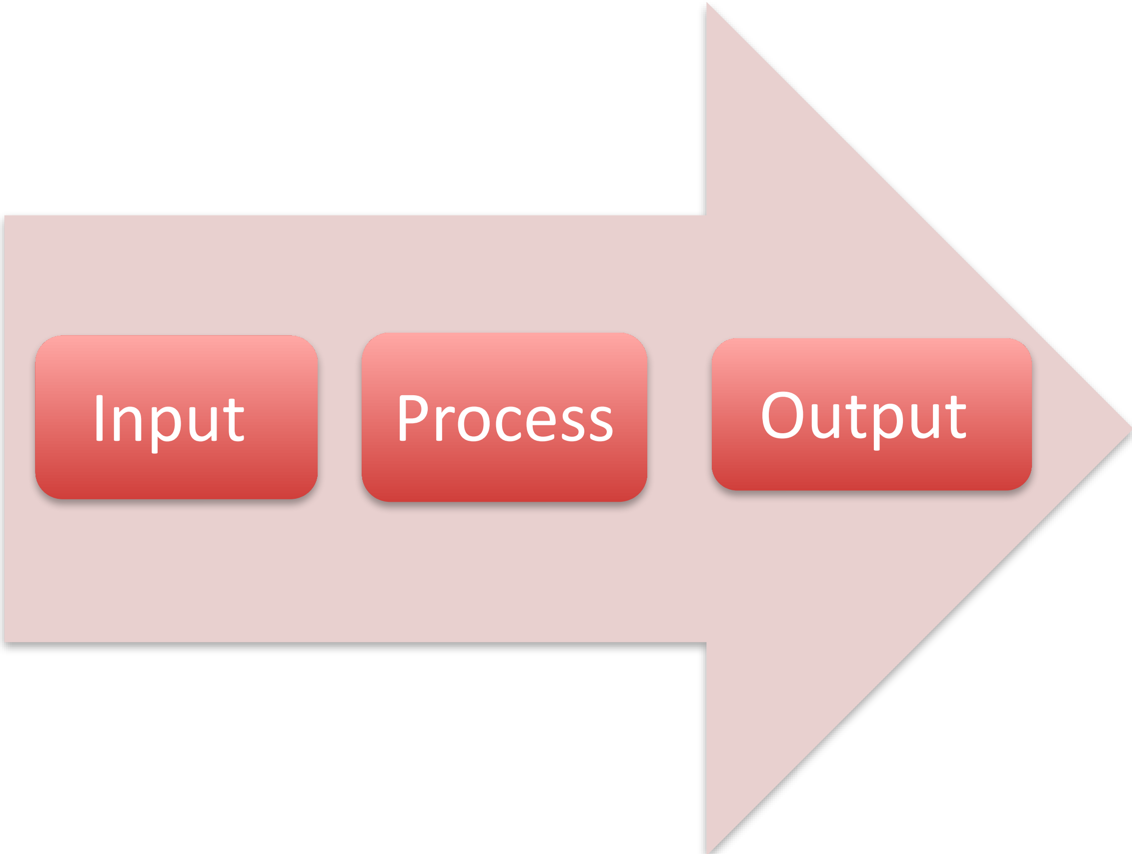
CHAPTER ONE
BASICS

- PC Concept
- Lab procedure and Tools
- Static Energy and Computer
- Safety Rules
- Preventing Maintenance and troubleshoot

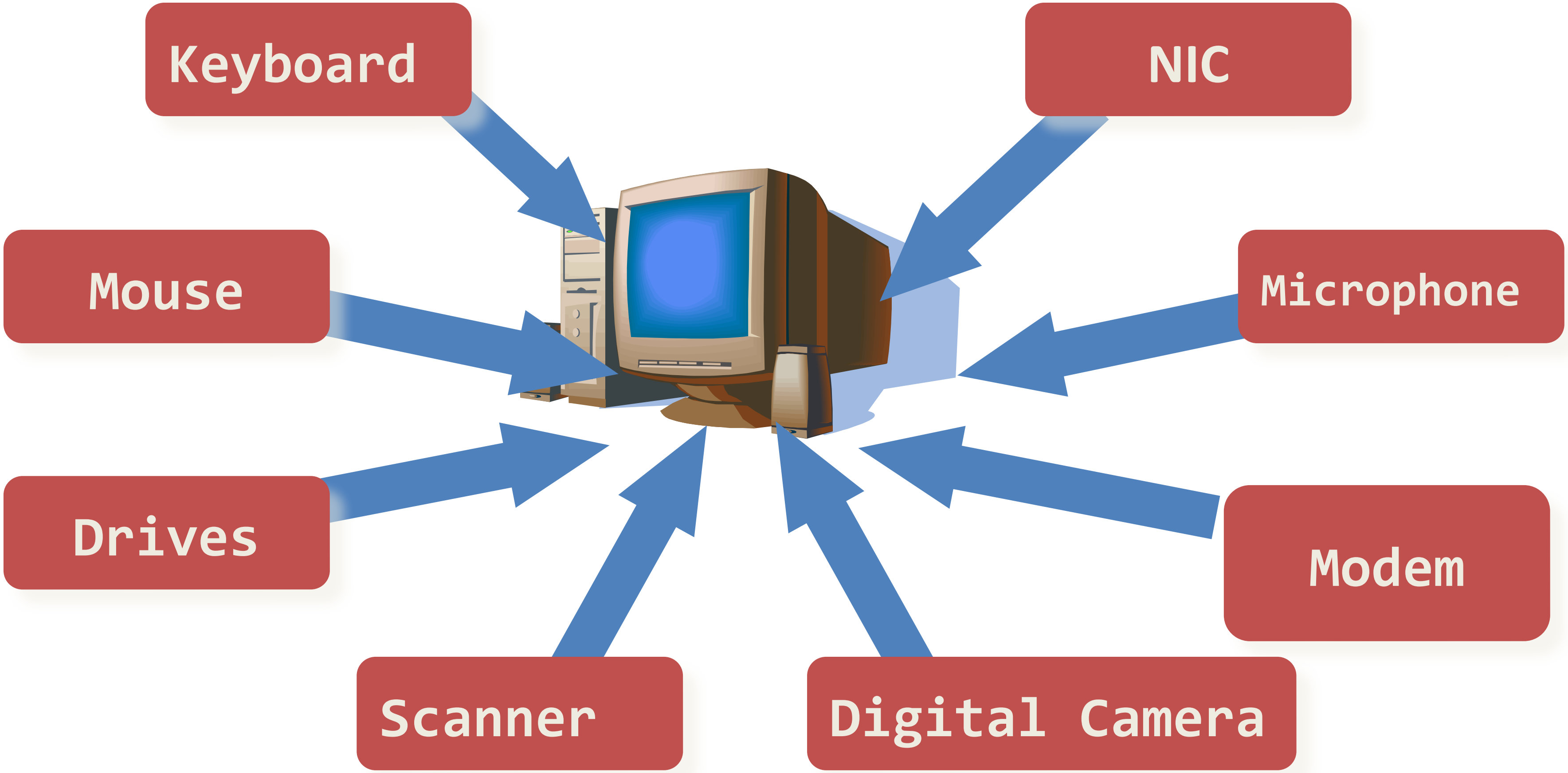
Computer


Computer
Hard Ware
Software
Input
Output
Processing Devices
Storage
Memory
System
Application

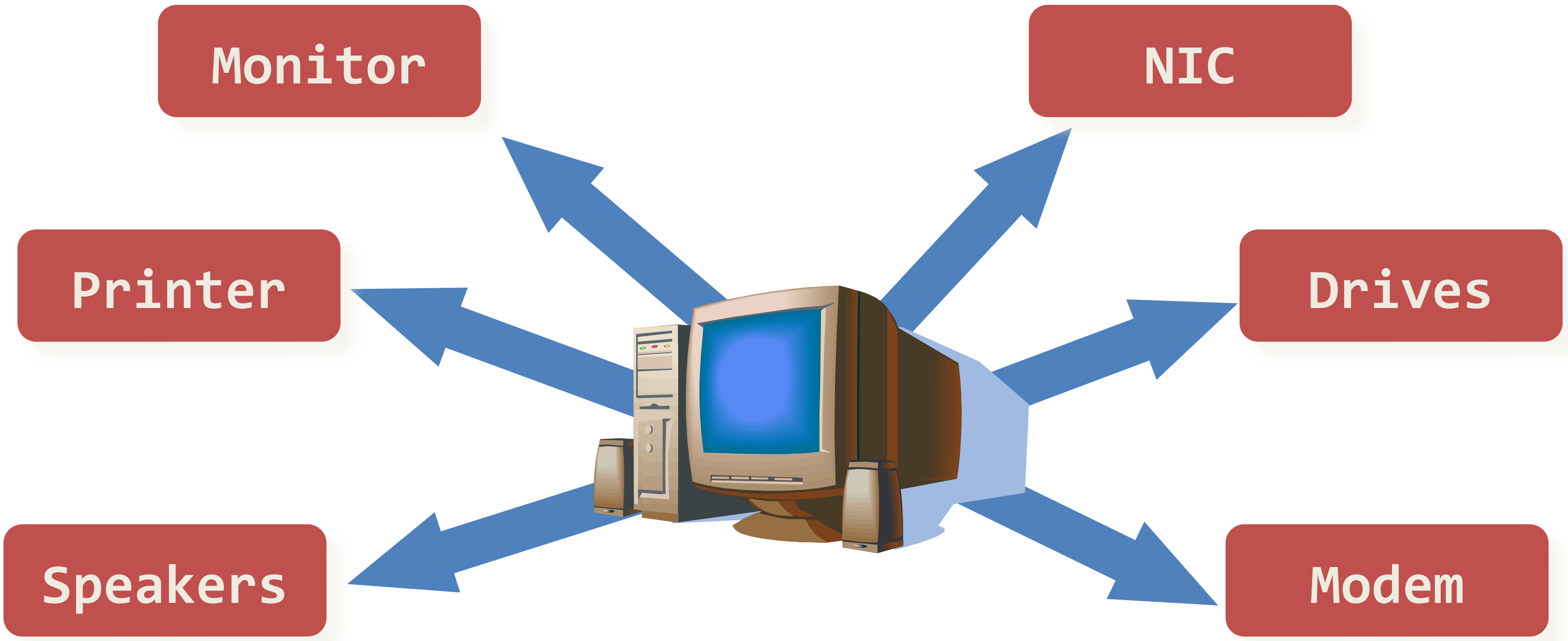
Computer



Input

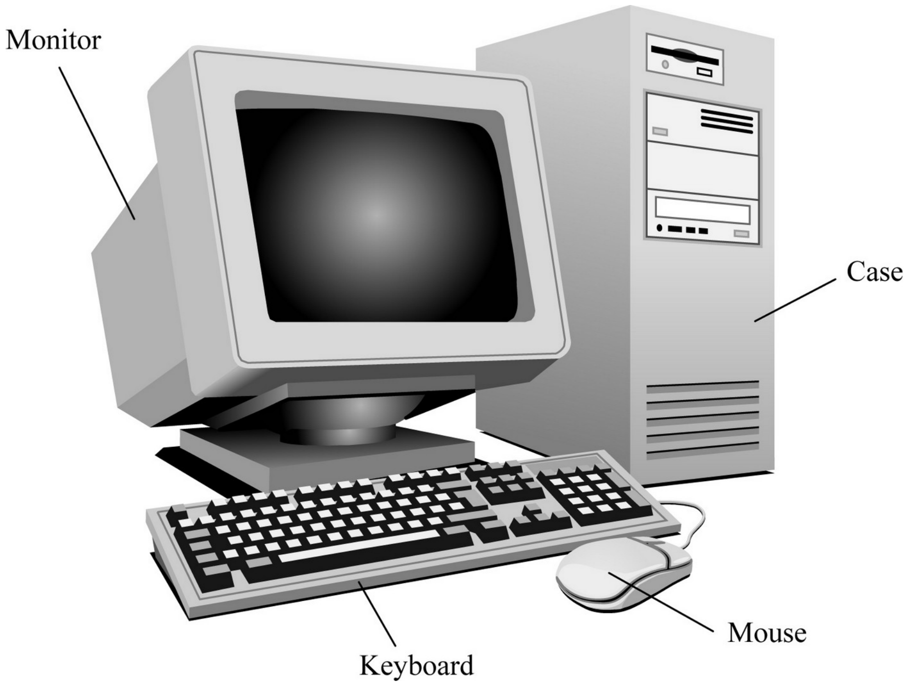

Keyboard
NIC
Mouse
Microphone
Drives
Modem
Scanner
Digital Camera

Output



Basic Computer Parts

Tower Computer

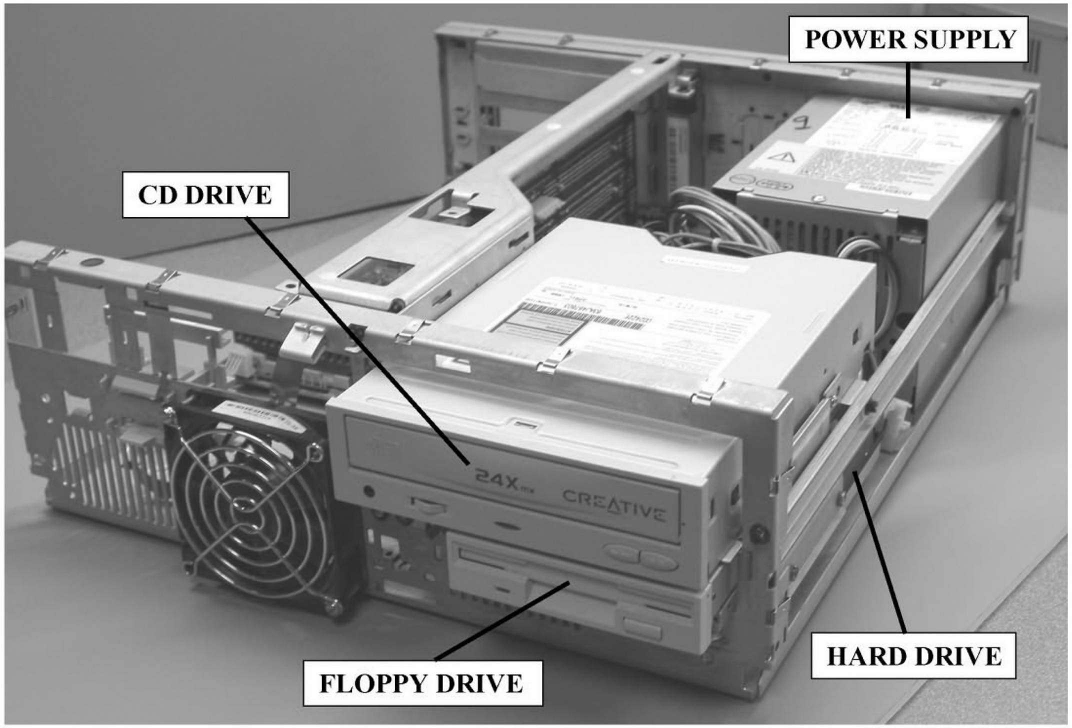


Basic Computer Parts

- Internal components of the computer include:
  - **power supply** - Converts AC voltage from the wall outlet to DC voltage the computer can use, supplies DC voltages for internal computer components and has **a fan to keep the computer cool.**
  - **floppy drive** - Common storage device that allows data storage to **floppy disks** (storage media) which can be used in other computers.
  - **hard drive** - Or hard disk, is a common storage device for maintaining files inside the computer, usually mounted below or beside the floppy drive.
  - **CD drive** - Holds disks (CDs) that have data, music, or software applications.
  - **DVD (Digital Versatile Disk) drive** - Popular alternative to a CD drive that supports CDs as well as music and video DVDs.

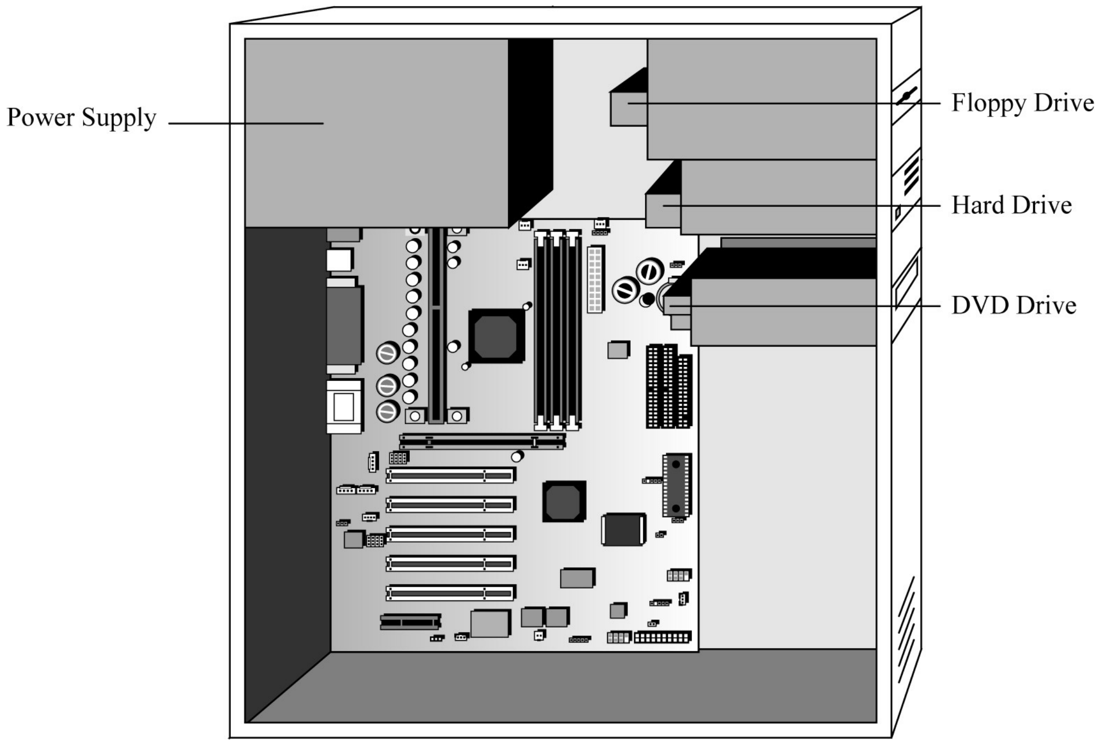
Basic Computer Parts

Desktop Computer with Hard Drive, Floppy Drive, CD Drive, and Power Supply



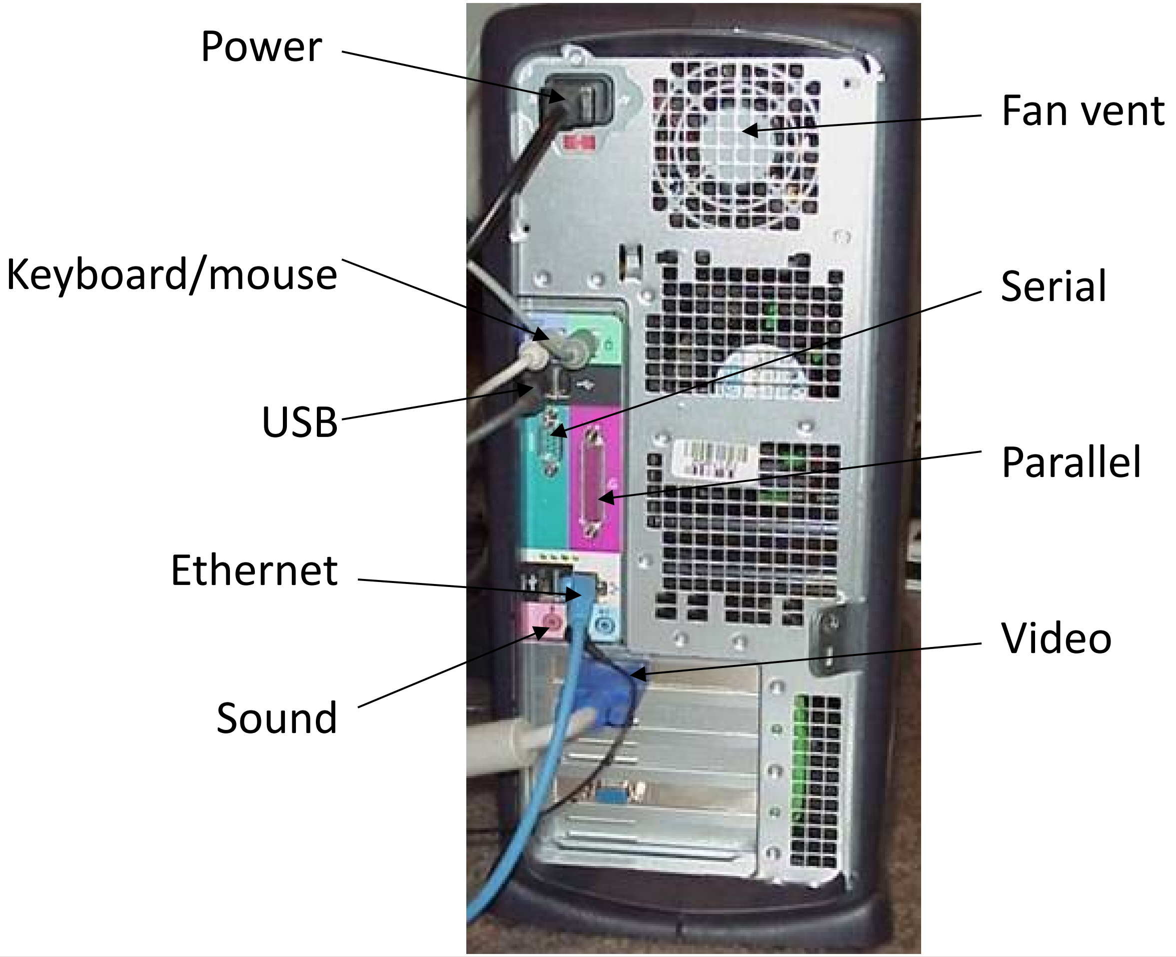
Basic Computer Parts

Tower Computer with Hard Drive, Floppy Drive, DVD Drive, and Power Supply



PC Exterior: Front

PC Exterior: Back



Basic Computer Parts

- More internal components of the computer are:
  - **Motherboard :**
    - The main circuit board that contains most of the electronics and is the largest electronic circuit board in the computer
    - all computer components connect to, or communicate through, the motherboard.
  - **Adapters :**
    - Smaller **electronic circuit cards** that normally plug into an ***expansion slot*** on the motherboard
    - allowing other devices to interface with the motherboard, they also may control some devices.

Notebook PCs

Built-in LCD screen

Built-in pointing device

Can run on AC or battery

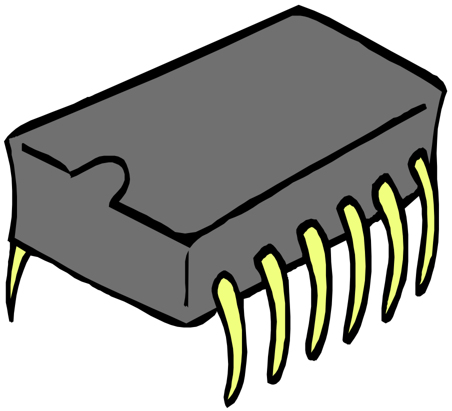
PCMCIA slots instead of ISA/PCI

Personal Digital Assistant (PDA)

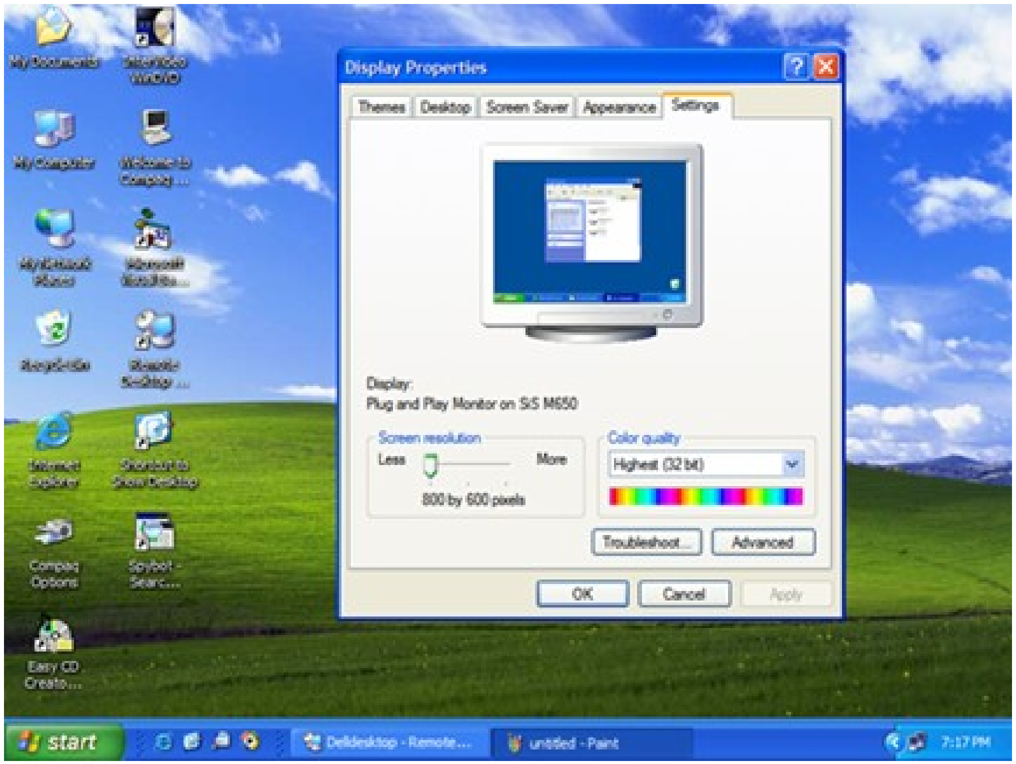
- ☑ Hand-held computer
- ☑ Limited amount of RAM
- ☑ Limited in functionality
- ☑ Write, tap with stylus on touch-sensitive screen

Basic Input Output System (BIOS)

- ☑ Startup instructions for low-level hardware
- ☑ Typically on a chip on the motherboard
- ☑ Does not change readily; requires special utility



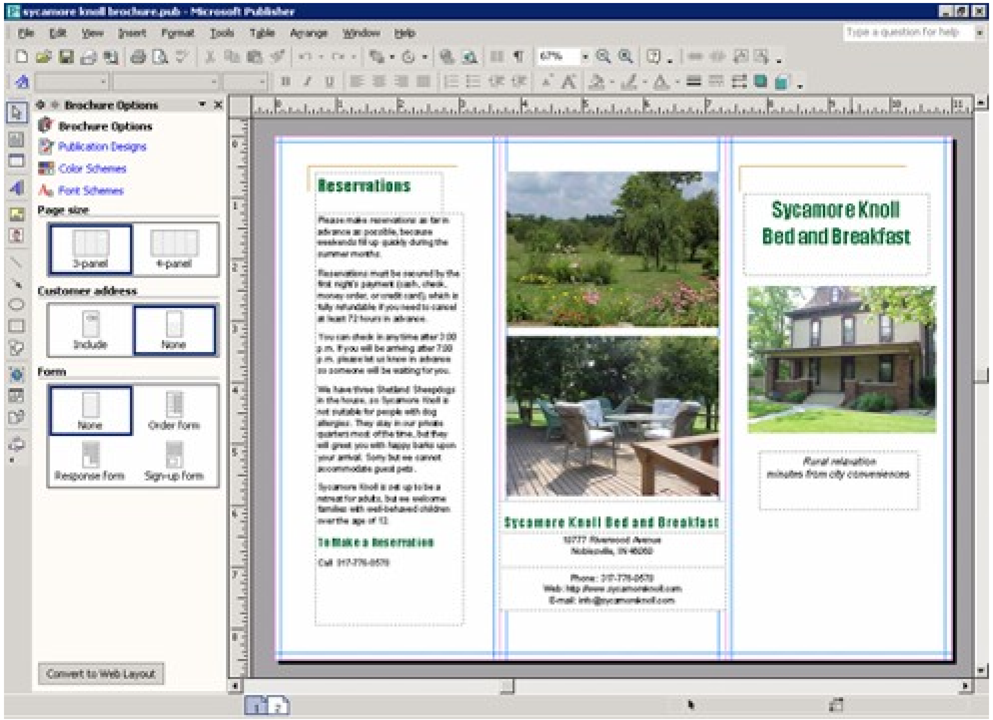
Operating System



- Interacts with human user
- Manages communication with software
- Runs applications
- Controls input and output

Applications

- Perform useful human tasks
- Run on top of an operating system
- Are written for a specific OS
- Examples: Word, Quicken, games



Safe Lab Procedures and Tool Use

- The workplace should have safety guidelines to follow to:
  - Protect people from injury
  - Protect equipment from damage
  - Protect the environment from contamination

General Safety Guidelines

- Most companies require reporting any injuries, including description of safety procedures not followed.
- Damage to equipment may result in claims for damages from the customer.

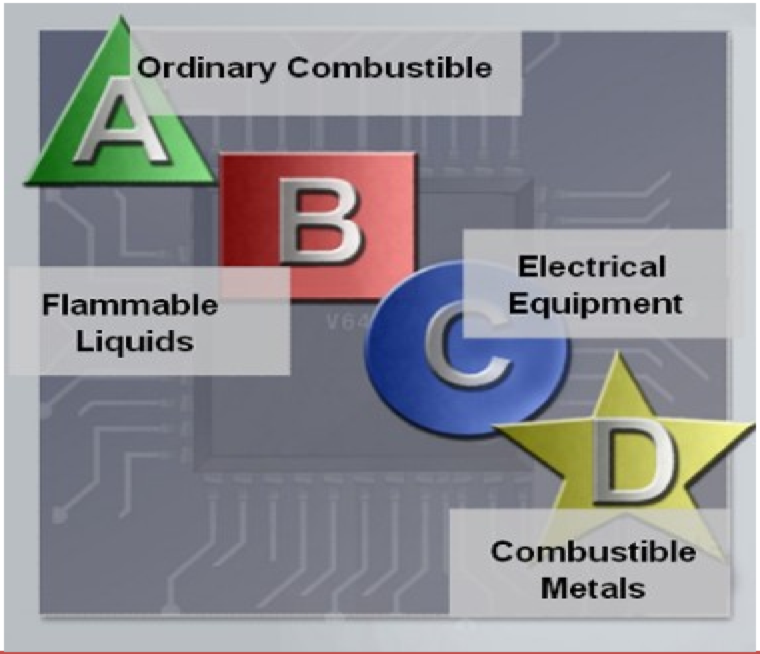
General Safety Guidelines

- Remove your watch or any other jewelry and secure loose clothing.
- Turn off the power and unplug equipment before performing service.
- Cover any sharp edges inside the computer case with tape.
- Never open a power supply or a monitor.
- Do not touch areas in printers that are hot or that use high voltage.
- Know where the fire extinguisher is located and how to use it.
- Keep food and drinks out of your workspace.
- Keep your workspace clean and free of clutter.
- Bend your knees when lifting heavy objects to avoid injury to your back.

Fire Safety Guidelines

Have a fire plan before you begin work:

- ☑ Know the location of fire extinguishers, how to use them, and which to use for electrical fires and for combustible fires
- ☑ Have an escape route in case a fire gets out of control
- ☑ Know how to contact emergency services quickly
- ☑ Keep the workspace clean
- ☑ Keep most solvents in a separate area



Power Fluctuation

AC power fluctuations can cause data loss or hardware failure:

- ☑ Blackouts, brownouts, noise, spikes, power surges

To help shield against power fluctuation issues, use power protection devices to protect the data and computer equipment:

- ☑ Surge suppressors
- ☑ UPS
- ☑ SPS

CAUTION: Never plug a printer into a UPS device. UPS manufacturers suggest not plugging a printer into a UPS for fear of burning up the printer motor.

Proper Disposal

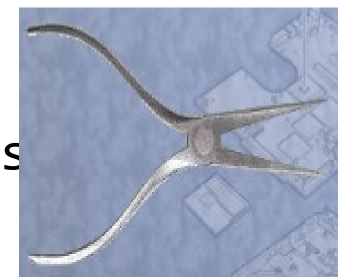
- Batteries from portable computer systems may contain lead, cadmium, lithium, alkaline manganese, and mercury.
- Recycling batteries should be a standard practice for a technician.

- Monitors contain up to 4 pounds of lead, as well as rare earth metals. Monitors must be disposed of in compliance with environmental regulations.
- Used printer toner kits and printer cartridges must be disposed of properly or recycled.
- Contact the local sanitation company to learn how and where to dispose of the chemicals and solvents used to clean computers.

Tools for the Job

Skilled use of tools and software makes the job less difficult and ensures that tasks are performed properly and safely.

- ESD Tools
  - antistatic wrist strap, mat
- Hand Tools
  - screwdrivers, needle-nose pliers
- Cleaning Tools
  - soft cloth, compressed air can
- Diagnostic Tools
  - digital multimeter,loopback adapter



Software Tools: Disk management tools

- **Fdisk**
  - create and delete disk partitions
- **Format**
  - prepare a hard drive prior to use
- **Scandisk or Chkdsk**
  - check for physical errors on the disk surface
- **Defrag**
  - optimize use of space on a disk
- **Disk Cleanup**
  - remove unused files
- **Disk Management**
  - creates partitions and formats disks (GUI interface)
- **System File Checker (SFC)**
  - scans the operating system critical files and replaces any files that are corrupt

Protection Software Tools

- ☑ **Windows XP Security Center** – allows you to check the status of essential security settings on the computer.
- ☑ **Antivirus Program** – protects a computer against virus attacks.
- ☑ **Spyware Remover** – protects against software that sends information about web surfing habits to an attacker. Spyware can be installed without the knowledge or consent of the user.
- ☑ **Firewall** – a program that runs continuously to protect against unauthorized communications to and from your computer.

Organizational Tools

- Personal reference tools
  - Notes, journal, history of repairs
- Internet reference tools
  - Search engines, news groups, manufacturer FAQs, online computer manuals, online forums and chats, technical websites
- Miscellaneous tools
  - Spare parts, a working laptop

Proper Use of Hand Tools

- ☑ Use the proper type and size of screwdriver by matching it to the screw.
  - ☑ Phillips, Flat Head and Hex are the most common types.
- ☑ Do not over tighten screws because the threads may become stripped.
- ☑ **Caution:** If excessive force is needed to remove or add a component, something may be wrong.
- ☑ **Caution:** Magnetized tools should not be used around electronic devices.
- ☑ **Caution:** Pencils should not be used inside the computer because the pencil lead can act as a conductor and may damage the computer components.

Proper Use of Cleaning Materials

To clean computers and accessories:

- ☑ Use mild cleaning solution and lint-free cloth to clean computer cases, outside of monitor, LCD screen, CRT screen, and mouse.
- ☑ Use compressed air to clean heat sinks.
- ☑ Use Isopropyl alcohol and lint-free swabs to clean RAM.
- ☑ Use hand-held vacuum cleaner with a brush attachment to clean a keyboard.

- ☑ **CAUTION:** Before cleaning any device, turn it off and unplug the device from the power source.

Preventive Maintenance
and
Troubleshooting

- ☑ Preventive Maintenance
- ☑ Troubleshooting

Preventive Maintenance

- ☑ **Preventive maintenance** is a regular and systematic **inspection**, **cleaning**, and **replacement** of worn parts, materials, and systems.
- ☑ Purpose of preventive maintenance
  - ☑ **Reduce** the likelihood of hardware or software **problems** by **systematically and periodically** checking hardware and software to ensure proper operation.
  - ☑ **Reduce** computer down time and repair **costs_**

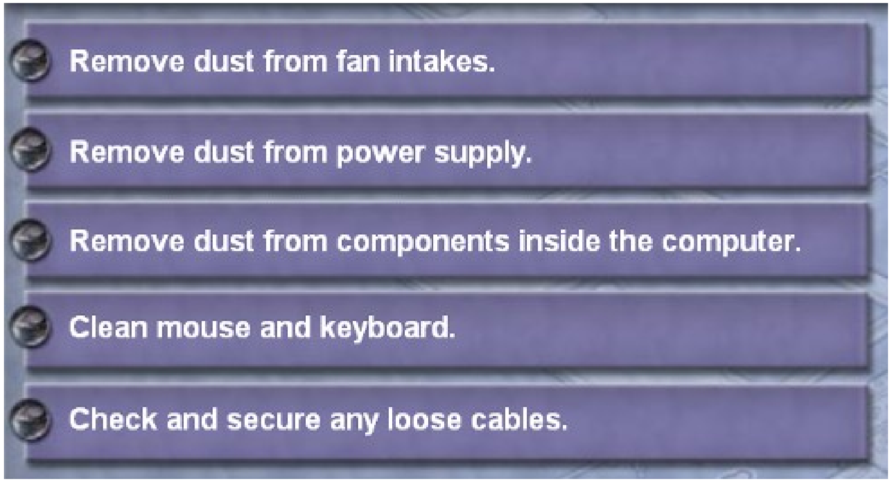
Troubleshooting

- ☑ **Troubleshooting** is a systematic approach to locating the cause of a fault in a computer system.
- ☑ Troubleshooting is a learned skill.
- ☑ Not all troubleshooting processes are the same
- ☑ technicians tend to refine their own **troubleshooting skills** based **on knowledge & personal experience.**

Hardware Maintenance

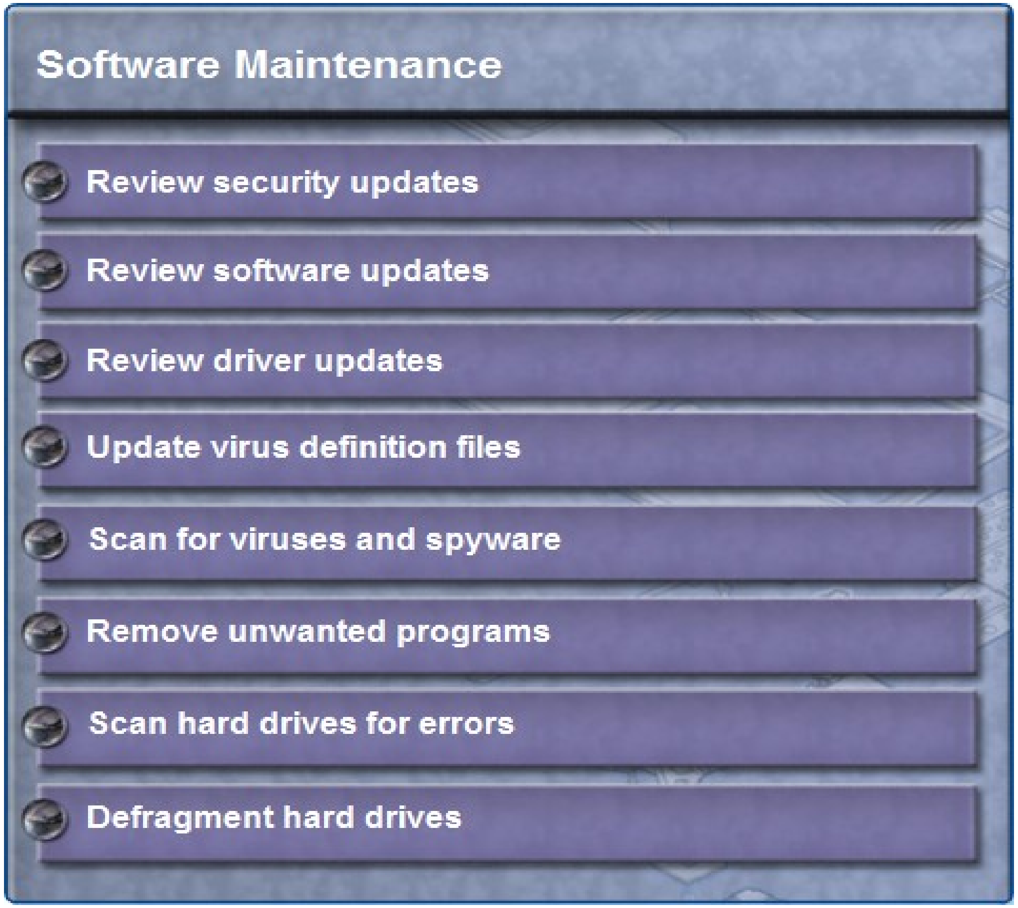
Make sure that the hardware is operating properly.

- ☑ Check the condition of parts.
- ☑ Repair or replace worn parts.
- ☑ Keep components clean.
- ☑ Create a hardware maintenance program.



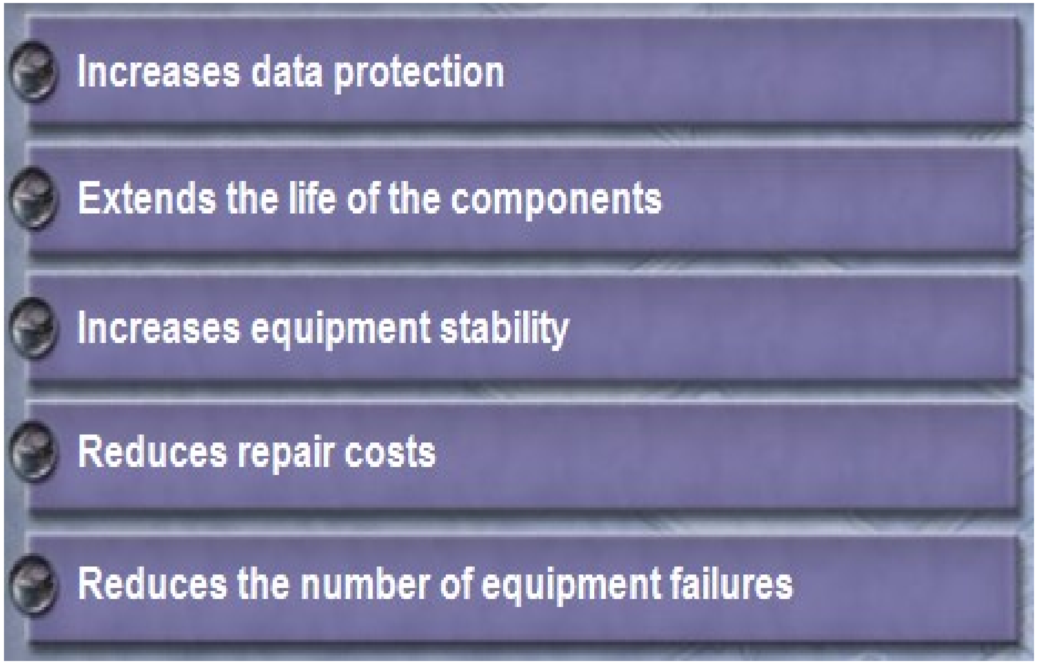
Software Maintenance

- ☑ Review updates
- ☑ Follow policies of your organization
- ☑ Create a schedule

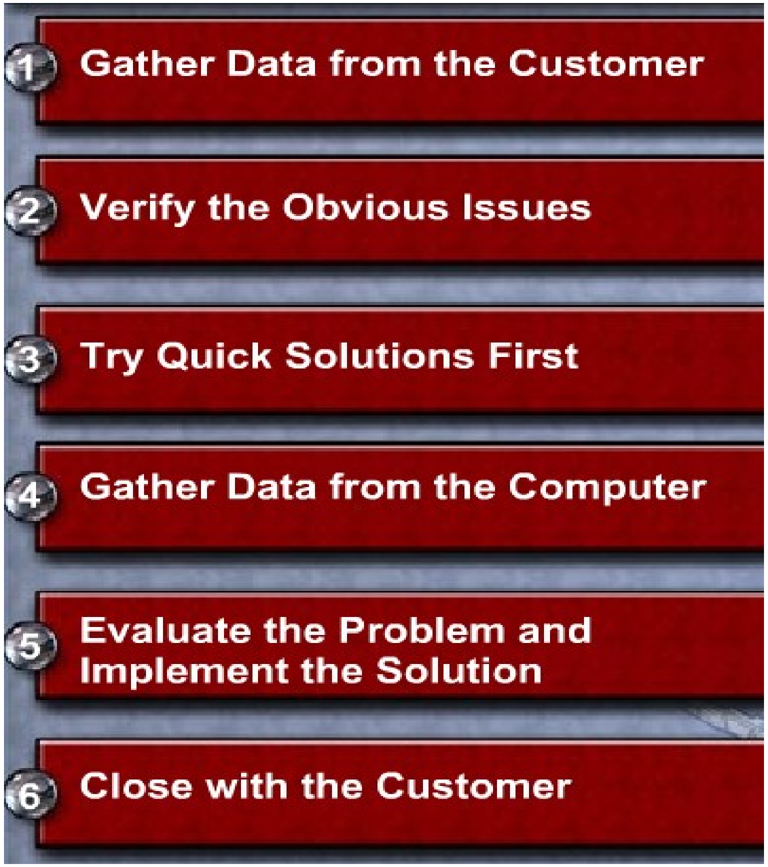


Preventive Maintenance Benefits

- ☑ Reduce computer down time.
- ☑ Reduce repair costs.
- ☑ Reduce loss of worker productivity.



The Troubleshooting Process



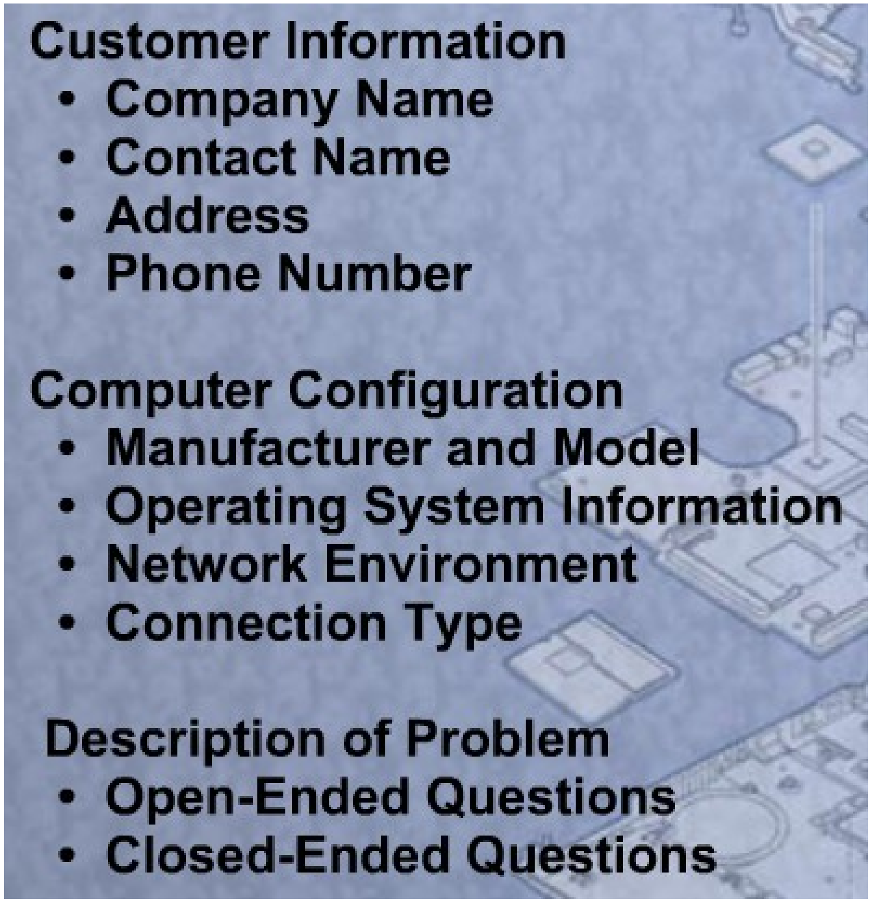
- ☑ Follow an organized and logical procedure.
- ☑ Address possible issues one at a time.
- ☑ Troubleshooting is a skill that is refined over time.
- ☑ The first and last steps involve effectively communicating with the customer.

Data Protection

Check with customer

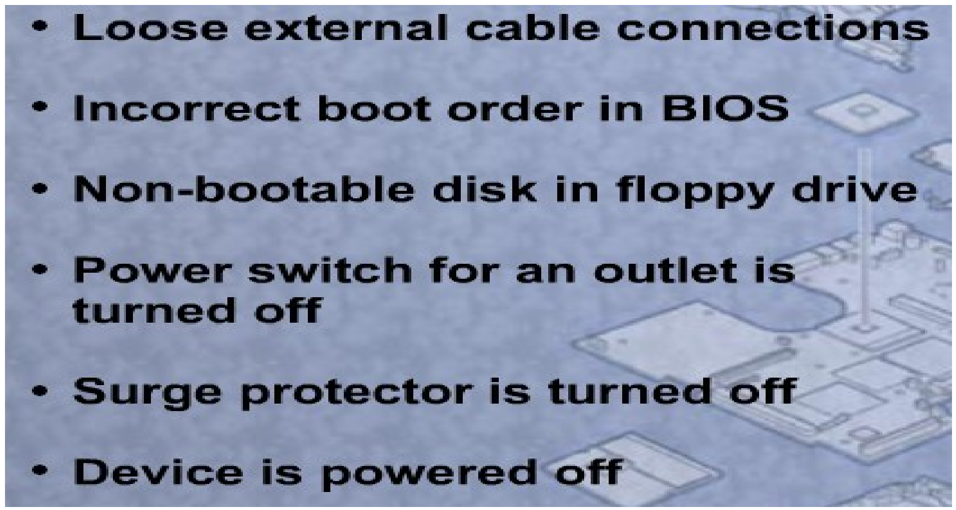
- ☑ Date of the last backup
- ☑ Contents of the backup
- ☑ Data integrity of the backup
- ☑ Availability of media for data restore If no backup can be created, ask customer to sign a release form

Gather Data from the Customer



- ☑ Communicate respectfully with the customer
- ☑ Start with open-ended questions
  - ☑ "What types of problems are you having with your computer or network?"
- ☑ Then, ask closed-ended (yes/no) questions
  - ☑ "Have you changed your password recently?"

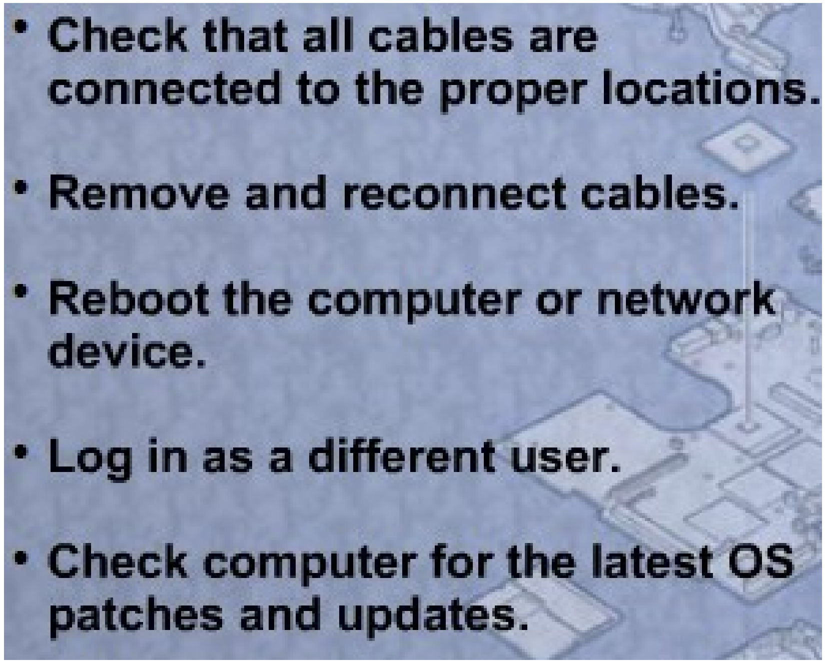
Verify Obvious Issues



- ☑ Problem may be simpler than the customer thinks.
- ☑ Checking for obvious issues can save time.
- ☑ If this step turns up nothing, continue to the next step of the troubleshooting process.

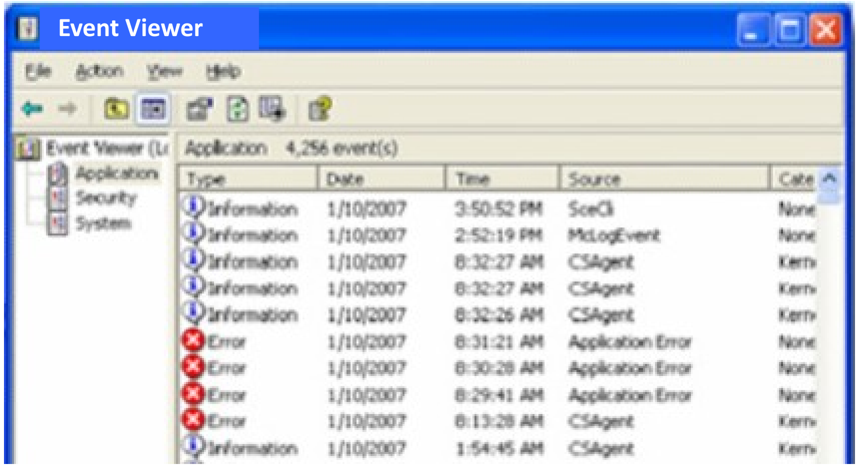
Try Quick Solutions

- May provide additional information, even if they do not solve the problem.
- Document each solution you try.
- May need to gather more information from the customer.
- If you find the problem at this stage, document it and proceed to the end of the troubleshooting process.



Gather Data from the Computer

- When system, user, or software errors occur on a computer, the **Event Viewer** is updated with information about the errors:
  - What problem occurred
  - The date and time of the problem
  - The severity of the problem
  - The source of the problem
  - Event ID number
  - Which user was logged in when the problem occurred

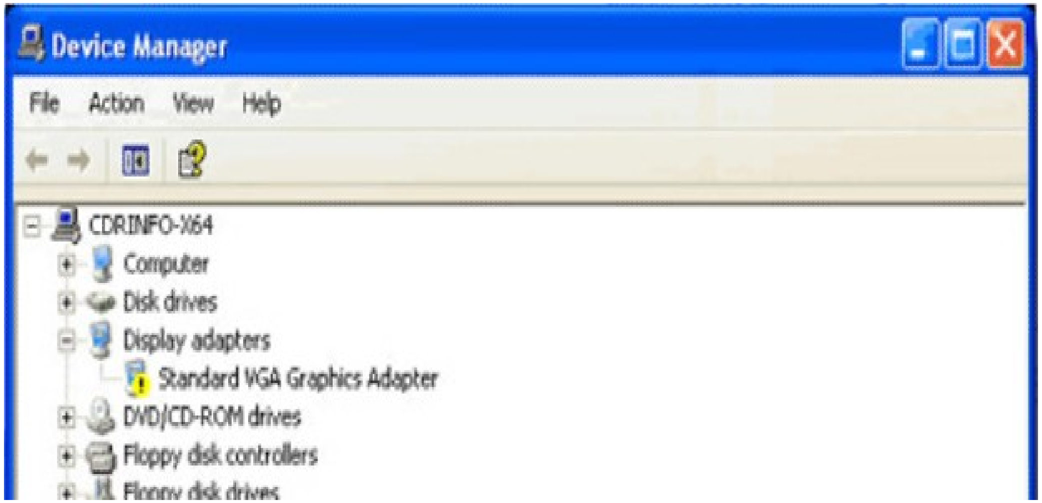


- Although this utility lists details about the error, you may still need to research the solution.

Gather Data from the Computer

Device Manager

- ☑ A flag of ! indicates the device is acting incorrectly.
- ☑ A flag of X indicates the device is disabled.



Gather Data from the Computer

- When troubleshooting, power on the computer and listen to the **beep code sequence**. Document the beep code sequence and research the code to determine the specific hardware failure.
- If the computer boots and stops after the POST, investigate the **BIOS settings** to determine where to find the problem. Refer to the motherboard manual to make sure that the BIOS settings are accurate.
- Conduct research to find software to use to diagnose and solve problems. Often, manufacturers of system hardware provide **diagnostic tools** of their own.

Evaluate the Problem, Implement the Solution

- Research possible solutions:

- Prioritize solutions to try.
- Try easiest solutions first.
- After an unsuccessful try, undo any changes you have made.
  - Unnecessary changes could complicate finding the solution.

Close with the Customer

- Discuss the solution with the customer
- Have the customer confirm that the problem has been solved
- Document the process
  - Problem description
  - Solution
  - Components used
  - Amount of time spent in solving the problem

Completed Work Order

The gratifying results of a day's work

Company Name: Cisco Systems, Inc.
Contact: Office Manager
Company Address: 170 West Tasman Drive, San Jose, CA 95134
Company Phone: 408-526-4000

Work Order

Generating a New Ticket

GENERATING A NEW TICKET

Category: HW
Type: Laptop
Item: Laptop
Closure Code:
Escalated? N
Status: CLOSED
Pending:
Pending Until Date:
Business Impacting? Yes No
Summary: Won't Boot
Case ID#: Cisco001
Priority: Medium
Connection Type: Wireless network connection
Environment: Mobile
User Platform: Windows XP

Problem Description:

User complains that the laptop won't boot up.
No software was added recently. No operating system changes have been made.
No peripherals have been added.

Problem Solution:

The existing battery is three years old. The exisiting battery does not boot the laptop even when the AC adapter is connected. A known good battery was installed and the laptop booted and passed all POST tests. The old battery appears to have a short in it that is causing the problem. The customer was advised that the old battery will need to be replaced, preferable with a new one. Time to complete repairs: 30 minutes.

END OF THE CHAPTER

Thank you